

# Lab1 solution

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## Part 1: Capturing HTTP Traffic

**Task 1: Start Wireshark and capture packets.**

**Task 2: Filter HTTP packets and analyze them.**

| http |              |                        |                        |          |        |
|------|--------------|------------------------|------------------------|----------|--------|
| No.  | Time         | Source                 | Destination            | Protocol | Length |
| 633  | 73.933542900 | 2001:16a2:3082:600:... | 2a03:90c0:6a1:2801:... | HTTP     |        |
| 636  | 74.137055200 | 2a03:90c0:6a1:2801:... | 2001:16a2:3082:600:... | HTTP     |        |
| 637  | 74.151780500 | 2001:16a2:3082:600:... | 2a03:90c0:6a1:2801:... | HTTP     |        |
| 639  | 74.341855700 | 2a03:90c0:6a1:2801:... | 2001:16a2:3082:600:... | HTTP     |        |

```
▼ Hypertext Transfer Protocol
  ▶ GET / HTTP/1.1\r\n
    Host: neverssl.com\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: ar,en;q=0.9,en-GB;q=0.8,en-US;q=0.7,ar-SA;q=0.6\r\n
    \r\n
    [Response in frame: 36200]
    [Full request URI: http://neverssl.com/]
```

## Part 2: Analyzing TCP/IP Traffic

### Task 1: Filter TCP packets

The image shows the Wireshark interface with a packet list on the left and packet details on the right. The packet list shows a series of TCP packets. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. A context menu is open over the packet details, showing options like 'Mark/Unmark Selected', 'Ignore/Unignore Selected', 'Set/Unset Time Reference', 'Time Shift...', 'Packet Comments', 'Edit Resolved Name', 'Apply as Filter', 'Prepare as Filter', 'Conversation Filter', 'Colorize Conversation', 'SCTP', and 'Follow'. The 'Follow' option is highlighted, and the 'TCP Stream' pane is visible on the right.

The image shows the Wireshark packet details pane for a selected TCP stream. The pane displays the raw data of the stream, which appears to be a series of characters. The pane also shows the packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet details pane is set to 'Follow TCP Stream' and shows the stream's data in ASCII format. The stream data is displayed in a text area, and the packet details pane shows the packet's structure and the stream's data.

### Task 2: Analyze TCP handshake and investigate Data Transfer and Termination

The image shows the Wireshark interface with a packet list on the left and packet details on the right. The packet list shows a series of TCP packets, including a SYN exchange and a data transfer. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet details pane is set to 'Follow TCP Stream' and shows the stream's data in ASCII format. The stream data is displayed in a text area, and the packet details pane shows the packet's structure and the stream's data.

```
74 80 → 27292 [FIN, ACK] Seq=2525 Ack=1558 Win=29160 Len=0
74 27292 → 80 [ACK] Seq=1558 Ack=2526 Win=65119 Len=0
74 27292 → 80 [FIN, ACK] Seq=1558 Ack=2526 Win=65119 Len=0
74 80 → 27292 [ACK] Seq=2526 Ack=1559 Win=29160 Len=0
```

## Part 3: Capturing and Analyzing UDP Traffic

```
250 5.690602400 2001:16a0:102:112::... 2001:16a2:3082:600::... UDP 1296 443 → 60615 Len=1230
251 5.690607500 2001:16a0:102:112::... 2001:16a2:3082:600::... UDP 1296 443 → 60615 Len=1230
252 5.690610200 2001:16a0:102:112::... 2001:16a2:3082:600::... UDP 1296 443 → 60615 Len=1230
▶ Frame 251: Packet, 1296 bytes on wire (10368 bits), 1296 bytes captured (10368 bits) on interface \Device\NPF_{15
▶ Ethernet II, Src: HuaweiTechno_77:8b:9e (f4:b7:8d:77:8b:9e), Dst: Intel_58:a5:43 (48:68:4a:58:a5:43)
▶ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
▶ Internet Protocol Version 6, Src: 2001:16a0:102:112::12, Dst: 2001:16a2:3082:600:f077:7484:40aa:b88b
▼ User Datagram Protocol, Src Port: 443, Dst Port: 60615
    Source Port: 443
    Destination Port: 60615
    Length: 1238
    Checksum: 0xb9de [unverified]
    [Checksum Status: Unverified]
    [Stream index: 4]
    [Stream Packet Number: 211]
    ▶ [Timestamps]
    UDP payload (1230 bytes)
▶ Data (1230 bytes)
```

UDP has a simpler header than TCP because it contains fewer fields and has lower overhead.

## Part 4: Compare TCP and UDP

### Task 1 :— Reliability and Connection Establishment

| Feature                  | TCP                                                       | UDP                                     |
|--------------------------|-----------------------------------------------------------|-----------------------------------------|
| Reliability              | Reliable (ensures delivery and retransmits lost packets). | Unreliable (no guarantee of delivery).  |
| Connection Establishment | Connection oriented (uses a three-way handshake).         | Connectionless (no handshake required). |

TCP provides reliable and ordered delivery, while UDP sends packets without establishing a connection or guaranteeing delivery.

### Task 2 :— Use Cases and Performance

| Feature     | TCP                                 | UDP                                   |
|-------------|-------------------------------------|---------------------------------------|
| Use Cases   | Web browsing, email, file transfer. | Video streaming, VoIP, online gaming. |
| Performance | More overhead, but reliable.        | Faster and lower overhead.            |